

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Consider the following truth table:

1 / 1 point

P	Q	$P \vee Q$
0	0	a
0	1	b
1	0	c
1	1	d

What is the value of b ?

- ☒ 1
- ☐ 0

👍 Nice work

Correct.

2. Considering the following formula, which one is correct?

1 / 1 point

$((r \rightarrow s) \wedge ((q \wedge r) \rightarrow q)) \wedge (r \rightarrow (s \rightarrow r))$

- ☐ It is a tautology.
- ☐ It is a Contradiction.
- ☒ It is consistent
- ☐ It is a tautology and consistent

👍 Nice work

Correct, for $r = \text{True}$, $q = \text{True}$ and $s = \text{True}$ the formula is true.

3. Consider the following logical expression:

1 / 1 point

$$(P \wedge Q) \vee (\neg P \wedge R)$$

. Which of the following truth tables correctly represents this logical expression?

☐

1				
2	P	Q	R	$(P \wedge Q) \vee (\neg P \wedge R)$
3	=====			
4	T	T	T	T
5	T	T	F	T
6	T	F	T	T
7	T	F	F	F
8	F	T	T	T
9	F	T	F	F
10	F	F	T	T
11	F	F	F	F
12				

☐

1				
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12				

☒

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☐

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10	F	F	T	F
11	F	F	F	F
12				

👍 Nice work

Well done! You've correctly evaluated the logical expression for all combinations of truth values.

4. Which one of the following statements is true, assume x is a natural number and $P(x) = x$ is odd number, $Q(x) = x$ is prime number?

1 / 1 point

- ☐ $\forall x, Q(x) \rightarrow \neg P(x)$
- ☐ Both $\forall x, Q(x) \rightarrow P(x)$ and $\exists x, P(x) \rightarrow P(x)$ are correct.
- ☐ $\forall x, Q(x) \rightarrow P(x)$
- ☒ $\exists x, P(x) \rightarrow Q(x)$
- ☐ $\forall x, P(x) \rightarrow Q(x)$

👍 Nice work

Correct, 7 is odd and it is a prime.

5. Which one of the following statements is true, assume x is a natural number and $P(x) = x$ is odd number, $Q(x) = x$ is prime number?

1 / 1 point

- ☐ $\forall x, Q(x) \rightarrow \neg P(x)$
- ☐ Both $\forall x, Q(x) \rightarrow P(x)$ and $\exists x, P(x) \rightarrow P(x)$ are correct.
- ☒ $\exists x, P(x) \rightarrow Q(x)$
- ☐ $\forall x, Q(x) \rightarrow P(x)$
- ☐ $\forall x, P(x) \rightarrow Q(x)$

👍 Nice work

Correct, 7 is odd and it is a prime.